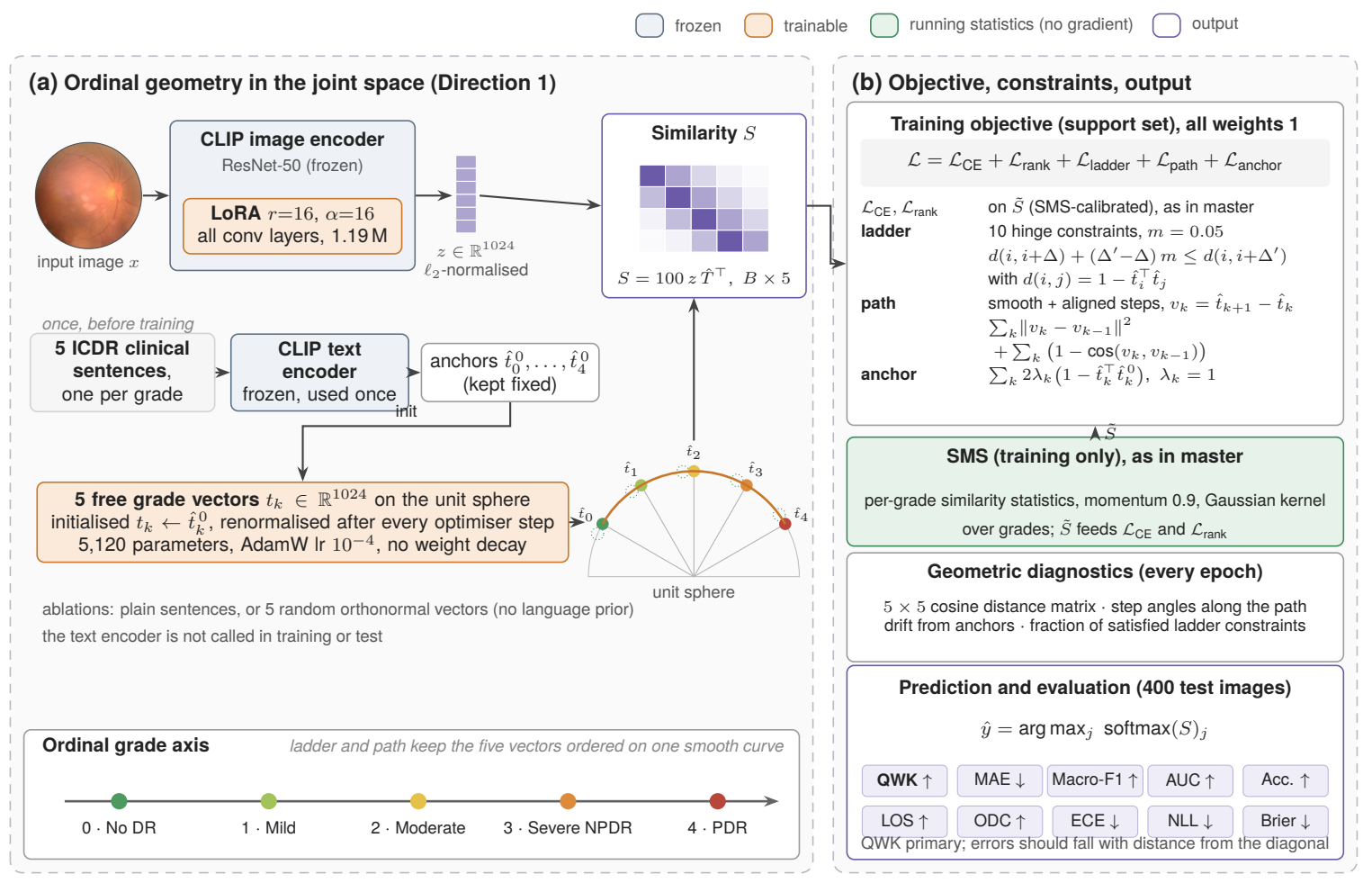
frozen
trainable
running statistics (no gradient)
output
(a) Ordinal geometry in the joint space (Direction 1)
input image x
CLIP image encoder
ResNet-50 (frozen)
LoRA r=16, α=16
all conv layers, 1.19 M
$z \in \mathbb{R}^{1024}$
ℓ_2-normalised
Similarity S
$S = 100\, z\, \hat{T}^\top,\ B \times 5$
once, before training
5 ICDR clinical sentences, one per grade
CLIP text encoder
frozen, used once
anchors $\hat{t}_0^0, \ldots, \hat{t}_4^0$
(kept fixed)
init
5 free grade vectors $t_k \in \mathbb{R}^{1024}$ on the unit sphere
initialised $t_k \leftarrow \hat{t}_k^0$, renormalised after every optimiser step
5,120 parameters, AdamW lr 10^{-4}, no weight decay
$\hat{t}_0$
$\hat{t}_1$
$\hat{t}_2$
$\hat{t}_3$
$\hat{t}_4$
unit sphere
ablations: plain sentences, or 5 random orthonormal vectors (no language prior)
the text encoder is not called in training or test
Ordinal grade axis
ladder and path keep the five vectors ordered on one smooth curve
0 · No DR
1 · Mild
2 · Moderate
3 · Severe NPDR
4 · PDR
(b) Objective, constraints, output
Training objective (support set), all weights 1
$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{rank}} + \mathcal{L}_{\text{ladder}} + \mathcal{L}_{\text{path}} + \mathcal{L}_{\text{anchor}}$
$\mathcal{L}_{\text{CE}}, \mathcal{L}_{\text{rank}}$ on $\tilde{S}$ (SMS-calibrated), as in master
ladder 10 hinge constraints, $m = 0.05$
$d(i, i+\Delta) + (\Delta' - \Delta)\, m \le d(i, i+\Delta')$
with $d(i,j) = 1 - \hat{t}_i^\top \hat{t}_j$
path smooth + aligned steps, $v_k = \hat{t}_{k+1} - \hat{t}_k$
$\sum_k \|v_k - v_{k-1}\|^2 + \sum_k (1 - \cos(v_k, v_{k-1}))$
anchor $\sum_k 2\lambda_k (1 - \hat{t}_k^\top \hat{t}_k^0),\ \lambda_k = 1$
$\tilde{S}$
SMS (training only), as in master
per-grade similarity statistics, momentum 0.9, Gaussian kernel over grades; $\tilde{S}$ feeds $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{rank}}$
Geometric diagnostics (every epoch)
5×5 cosine distance matrix · step angles along the path
drift from anchors · fraction of satisfied ladder constraints
Prediction and evaluation (400 test images)
$\hat{y} = \arg\max_j\ \text{softmax}(S)_j$
QWK ↑
MAE ↓
Macro-F1 ↑
AUC ↑
Acc. ↑
LOS ↑
ODC ↑
ECE ↓
NLL ↓
Brier ↓
QWK primary; errors should fall with distance from the diagonal